# BEYOND CRAMÉR: A UNIFIED POISSON LIMIT FRAMEWORK FOR PRIME-REPRESENTING SIEVE SPACINGS

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Abstract. We establish a unified geometric framework that reduces the probabilistic distribution of prime-representing sieves to deterministic modular intersections. By formalizing a class of bounded-variance, multi-residue sieves, we prove that Poisson-distributed spacings are a necessary consequence of local geometric correlation, bridging historically distinct configurations. This unified theory resolves the transition from sieves with constant dimensions, such as the twin prime sieve ($x(x+2)$), to sieves with fluctuating dimensions governed by the Legendre symbol, such as Landau's First Problem ($n^2+1$). Furthermore, we machine-verify a Grand Convergence Theorem, proving that any multi-residue sieve satisfying the collision threshold axioms admits a factorization of the $k$-tuple expectation into a single-event density and an absolutely convergent geometric correlation constant.

## 1. Introduction

The Cramér model [2] hypothesizes that primes behave as independent stochastic variables, providing heuristic justifications for a wide array of asymptotic distributions. While remarkably predictive, this model is a statistical assumption superimposed on a deterministic sequence. The apparent randomness of the primes arises not from independence but from the intersections of modular residue classes over bounded intervals. To make this precise, modern analytic number theory relies on multi-residue sieve frameworks [3], evaluating the surviving integers that avoid forbidden congruence classes across a primorial base $q_y = \prod_{p \le y} p$.

Historically, the application of multi-residue sieves to prime representation problems has proceeded along separate lines. Linear configurations, such as the Twin Prime Conjecture, have a constant sieve dimension. Conversely, quadratic configurations, such as Landau's First Problem[1] regarding primes of the form $n^2+1$, exhibit a fluctuating sieve dimension. In the quadratic case, primes $p \equiv 1 \pmod 4$ exclude two residue classes, primes $p \equiv 3 \pmod 4$ exclude none, and the distance between the roots modulo $p$ varies with $p$. This dimensional fluctuation has obstructed attempts to model these configurations under a single probabilistic limit.

This paper presents the "Beyond Cramér" framework: a unified theory showing that Poisson spacings follow from bounded collision structure in the sieve, independent of whether the underlying polynomial is linear or quadratic. We introduce

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[1]The list of four basic problems about prime numbers presented by Edmund Landau at the 1912 ICM [6] has been reported in various, non-consistent orders by several different sources. We choose to use the order in which they are listed in the actual proceedings of the congress.

an abstract axiomatic system —the Poisson-admissible sieve—characterized by a localized density map $\kappa(p)$ and a collision threshold.

We establish that if a sieve's local correlation deviation is bounded within $1 + O(1/p^2)$, the global $k$-tuple expectation factorizes. The main result, the Grand Convergence Theorem, shows that any sieve satisfying these axioms converges to a Poisson point process.

Our paper is structured as follows:

(1) **The Unified Sieve Axioms:** In Section 2, we outline the core structural requirements of a Poisson-admissible sieve, reducing the combinatorial variance to a tractable limit.
(2) **The Grand Convergence Theorem:** In Section 3, we provide the analytic proof that admissibility forces Poisson spacings, reducing the requirement of equidistribution to bounded local overlaps.
(3) **Geometric Realizations:** In Sections 4 and 5, we present the specializations for the twin prime and Landau $(n^2 + 1)$ sieves, respectively, proving that both these two configurations are instances of the same axiomatic framework.

Throughout this text, 卐 symbols serve as direct links to formalized code in the GitHub repository `github.com/ElNando888/BeyondCramer/tree/v1.2`. This repository contains the `sorry`-free[2] machine-checked Lean 4 verifications for the core definitions, the Grand Convergence factorization, and both geometric specializations discussed in this paper.

## 2. The Unified Sieve Axioms

To move beyond statistical heuristics, we isolate the geometric properties that govern the distribution of surviving residue classes. Whether a multi-residue sieve targets a linear configuration or a quadratic polynomial, its limiting behavior depends on its local density and the structure of its collision boundaries. We formalize these shared properties through the definition of a *Poisson-admissible sieve* [卐].

**Definition 2.1** (Poisson-Admissible Sieve)**.** A multi-residue sieve evaluated over the primorial $q_y = \prod_{p \le y} p$ is Poisson-admissible if it satisfies the following structural axioms:

(1) **Density Map $\kappa(p)$:** There exists a locally defined map $\kappa : \mathbb{P} \to \mathbb{N}$ representing the number of forbidden residue classes modulo $p$ generated by a single shift. This map must be bounded by a constant $\kappa(p) \le B$, and satisfy $\kappa(p) < p$ for all primes to ensure positive survival probabilities.
(2) **Collision Threshold $T(\mathbf{h})$:** For any fixed $k$-tuple of distinct integer shifts $\mathbf{h} = \{h_1, \ldots, h_k\}$, there exists a prime threshold $T(\mathbf{h})$ beyond which the excluded residue classes of distinct shifts are disjoint. For all $p > T(\mathbf{h})$, the total number of excluded classes is exactly $|S_p(\mathbf{h})| = \kappa(p) \cdot k$.
(3) **Geometric Convergence Bound:** The deviation of the local correlation factor from 1 is bounded. For all primes $p$ where $\kappa(p) \cdot k < p$, there exists a

[2]Running the `lake build` command will result in an output with a warning about a `sorry`, but this warning is related to the dependent project `PoissonViaCRT`, the formalization of the 2008 paper by Granville & Kurlberg [4], which while well-advanced, is still incomplete at the time of writing.

constant $C > 0$ such that:

$$\left| \frac{1 - \frac{\kappa(p) \cdot k}{p}}{\left(1 - \frac{\kappa(p)}{p}\right)^k} - 1 \right| \leq \frac{C}{p^2}$$

**Lemma 2.2** (Analytic Summability[3] [𑖌𑖽]). *For any Poisson-admissible sieve, the absolute deviation of the local correlation factor is summable over the primes.*

*Proof.* By the geometric convergence bound axiom, the deviation is strictly dominated by $C/p^2$ for all sufficiently large primes. Because the sum $\sum p^{-2}$ is absolutely convergent, the series of absolute deviations inherently converges by the direct comparison test. □

By extracting these axioms, we decouple the probabilistic limit of the sieve from the algebraic origins of its roots. The dimension of the sieve—whether constant as in linear configurations or fluctuating as in quadratic forms—is encoded in the density map $\kappa(p)$.

## 3. The Grand Convergence Theorem

With the combinatorial variance controlled by the axioms of Definition 2.1, we establish a factorization for the expected number of surviving shifted configurations.

**Definition 3.1** (Baseline Sieve Density [𑖌𑖽]). For an admissible sieve, the expected density of a single surviving element evaluated up to a parameter $y$ is governed by the density map $\kappa(p)$:

$$\rho(y) = \frac{1}{2} \prod_{2 < p \leq y} \left(1 - \frac{\kappa(p)}{p}\right)$$

The factor $1/2$ accounts for the exclusion at $p = 2$, which is treated separately from the product over odd primes.

In a purely independent stochastic model, the expected survival of a $k$-tuple would scale exactly as $\rho(y)^k$. However, the intersections of modular residue classes introduce local correlations. Let $\mathrm{excl}(p)$ denote the number of excluded classes modulo $p$ for the tuple $\mathbf{h}$, bounded by $\mathrm{excl}(p) \leq \kappa(p) \cdot k$.

**Theorem 3.2** (Unified Expectation Factorization [𑖌𑖽]). *For any Poisson-admissible sieve and any $k \geq 1$, the expected number of surviving $k$-tuples modulo the primorial $q_y$, denoted $E[N_k(\mathbf{h})]$, factors as the product of the structural interval, the independent stochastic probability, and a geometric correlation constant:*

$$E[N_k(\mathbf{h})] = q_y \cdot \left[\rho(y)\right]^k \cdot \mathfrak{G}(\mathbf{h})$$

*where the geometric correlation constant $\mathfrak{G}(\mathbf{h})$ is explicitly defined by the Euler product:*

$$\mathfrak{G}(\mathbf{h}) = \prod_{2 < p \leq y} \frac{1 - \frac{\mathrm{excl}(p)}{p}}{\left(1 - \frac{\kappa(p)}{p}\right)^k}$$

[3] In the formalized library, this summability is bundled as a class field for computational efficiency, but mathematically it is a direct consequence of the third axiom.

*Proof.* By definition, the expected number of surviving shifted $k$-tuples modulo the primorial $q_y$ evaluates the surviving density over each prime factor up to $y$:

$$E[N_k(\mathbf{h})] = q_y \cdot \left(\frac{1}{2}\right)^k \prod_{2<p\leq y} \left(1 - \frac{\mathrm{excl}(p)}{p}\right)$$

To factor out the independent stochastic probability $\rho(y)^k$, we multiply and divide the terms inside the product by the single-event survival probability raised to the $k$-th power, $(1-\kappa(p)/p)^k$. This is valid because the non-degeneracy axiom guarantees $\kappa(p) < p$, ensuring the denominator is positive.

Rearranging the terms algebraically yields:

$$E[N_k(\mathbf{h})] = q_y \cdot \left[\left(\frac{1}{2}\right)^k \prod_{2<p\leq y} \left(1 - \frac{\kappa(p)}{p}\right)^k\right] \cdot \prod_{2<p\leq y} \frac{1 - \frac{\mathrm{excl}(p)}{p}}{\left(1 - \frac{\kappa(p)}{p}\right)^k}$$

The bracketed term is $\rho(y)^k$ and the remaining product defines $\mathfrak{G}(\mathbf{h})$, establishing the factorization. □

While Theorem 3.2 algebraically untangles the expected yield, its usefulness depends on the behavior of $\mathfrak{G}(\mathbf{h})$ as the sieve parameter $y \to \infty$.

**Theorem 3.3** (Absolute Convergence of Geometric Correlation [𑖌]). *Because a Poisson-admissible sieve has a collision threshold $T(\mathbf{h})$ and a local deviation bounded by $O(1/p^2)$, the infinite product defining $\mathfrak{G}(\mathbf{h})$ is multipliable, converging absolutely to a positive, finite limit.*

*Proof.* To establish the absolute convergence of the infinite product defining $\mathfrak{G}(\mathbf{h})$ as $y \to \infty$, we analyze the local correlation factors:

$$L_p = \frac{1 - \frac{\mathrm{excl}(p)}{p}}{\left(1 - \frac{\kappa(p)}{p}\right)^k}$$

By the collision threshold axiom of the Poisson-admissible sieve, there exists a deterministic prime $T(\mathbf{h})$ such that for all primes $p > T(\mathbf{h})$, the excluded residue classes are disjoint, so $\mathrm{excl}(p) = \kappa(p) \cdot k$.

Substituting this maximal exclusion into the local factor for large primes yields $L_p = \frac{1-\kappa(p)k/p}{(1-\kappa(p)/p)^k}$. By the geometric convergence bound axiom, the deviation of this term from 1 satisfies $|L_p - 1| \leq C/p^2$ for some constant $C > 0$.

Because the sum over primes $\sum p^{-2}$ is absolutely convergent, the series of absolute deviations $\sum |L_p - 1|$ is also convergent. By the standard convergence criteria for Euler products (cf. [7, Chapter 1]), the absolute convergence of this series guarantees that the infinite product $\prod L_p$ converges to a positive, finite limit. □

Theorem 3.3 provides the quantitative link to the Granville–Kurlberg framework [4]. Because the deviations of the local correlation factor vanish at rate $O(1/p^2)$, the well-distribution hypothesis required for higher-order Poisson moments is satisfied. To formally bridge this abstract convergence to the spatial distribution of residues, we map the sieve into a concrete combinatorial structure.

**Definition 3.4** (Sieve Realization [𑖌]). A sieve realization equips a Poisson-admissible sieve with the specific topological data required to evaluate local spacings:

(1) A set of surviving residues $\Omega_p \subseteq \mathbb{Z}/p\mathbb{Z}$ for each prime $p$, which must be strictly non-empty.
(2) A well-distribution parameter $\varepsilon > 0$.
(3) A quantitative well-distribution bound satisfying Hypothesis 1 of Granville and Kurlberg [4], ensuring the local tuple counts deviate gracefully.
(4) A spacing bound controlling the maximum gap growth, $p/|\Omega_p| \leq p^{\lambda_K - \varepsilon}$.

**Corollary 3.5** (Unified Poisson Spacing [卐])**.** *For any Poisson-admissible sieve equipped with a valid sieve realization (Definition 3.4), the normalized spacings between surviving residues converge to a Poisson distribution. Specifically, for any test box $X$, the $k$-level correlation $R_k(X, \Omega_q)$ converges to the volume of the box,* $\mathrm{vol}(X)$*, as the average spacing between surviving elements tends to infinity.*

*Proof.* The Granville–Kurlberg framework for Poisson statistics in multi-residue sieves states that if a sequence of subsets satisfies a well-distribution hypothesis—namely, that the localized error terms vanish sufficiently fast—the $k$-level correlations converge to the volume of the test box.

For a Poisson-admissible sieve, the well-distribution hypothesis translates to the requirement that the deviations of the local correlation factors are analytically summable. By Theorem 3.3, the admissible sieve axioms guarantee $|L_p - 1| \leq C/p^2$. This inverse-square decay satisfies the error bounds required by the Granville–Kurlberg main theorem.

Consequently, the admissible sieve axioms imply the analytic requirements of well-distribution. As the average spacing $s_q$ between surviving elements tends to infinity, the correlation error $O(s_q^{-\delta})$ decays to zero, so the spacing distribution converges to a Poisson point process. □

This provides a structural explanation for the Cramér model. Poisson-distributed spacings among prime-representing polynomials are not an artifact of pseudorandomness but rather the analytic consequence of any sieve geometry with bounded local variance.

## 4. The Twin Prime Specialization (Constant Dimension)

The Twin Prime Conjecture, asserting the existence of infinitely many prime pairs $(p, p+2)$, is the canonical linear configuration. Evaluated as a multi-residue sieve, we target the polynomial $x(x+2)$, which excludes residue classes to prevent either factor from having a prime divisor. Because the polynomial is linear in its factors, the sieve dimension is constant across all primes $p > 2$.

**Definition 4.1** (Twin Prime Density Map [卐])**.** For the twin prime configuration, the density map $\kappa(p)$ counts the distinct roots of $x(x+2) \equiv 0 \pmod{p}$. Thus:

$$\kappa(p) = \begin{cases} 1 & \text{if } p = 2 \\ 2 & \text{if } p > 2 \end{cases}$$

**Lemma 4.2** (Linear Admissibility)**.** *The Twin Prime density map $\kappa(p)$ satisfies the boundedness and non-degeneracy axioms of a Poisson-admissible sieve.*

*Proof.* By Definition 4.1, the maximum value attained by the map is 2, providing the uniform bound $B = 2$. For non-degeneracy, we verify $\kappa(p) < p$ for all primes. For $p = 2$, $\kappa(2) = 1 < 2$. For all odd primes $p \geq 3$, $\kappa(p) = 2 < p$. Therefore, the survival probability at every prime is positive. □

A key feature of the twin prime sieve is its constant dimension. To verify the geometric correlation bound, we must show that local overlaps between distinct shifted configurations are bounded.

**Definition 4.3** (Linear Collision Threshold [ক্ষ])**.** For a fixed $k$-tuple of distinct integer shifts $\mathbf{h} = \{h_1, \ldots, h_k\}$, we define the prime threshold $T(\mathbf{h})$ as the maximum absolute difference between any interacting roots of the shifted polynomials across all pairs $(i, j)$ where $i \neq j$:

$$T(\mathbf{h}) = \max_{i \neq j} \big( |h_i - h_j|, |h_i - h_j - 2|, |h_i - h_j + 2| \big)$$

**Lemma 4.4** (Finiteness of Linear Overlaps)**.** *For all primes $p > T(\mathbf{h})$, the excluded residue classes generated by distinct shifts in the $k$-tuple are disjoint, yielding exactly $2k$ excluded classes modulo $p$.*

*Proof.* For a target coordinate $x$ shifted by $h_i$, the excluded residue classes modulo $p$ correspond to the roots $x \equiv -h_i \pmod{p}$ and $x \equiv -h_i - 2 \pmod{p}$. A collision between the excluded classes of two distinct shifts $h_i$ and $h_j$ occurs if and only if one of the following modular equivalences holds:

$$\begin{aligned} -h_i &\equiv -h_j \pmod{p} \implies p \mid (h_i - h_j) \\ -h_i &\equiv -h_j - 2 \pmod{p} \implies p \mid (h_i - h_j - 2) \\ -h_i - 2 &\equiv -h_j \pmod{p} \implies p \mid (h_i - h_j + 2) \end{aligned}$$

By Definition 4.3, the threshold $T(\mathbf{h})$ bounds the absolute value of all these differences. Because the shifts are distinct integers, these differences are non-zero. For any prime $p > T(\mathbf{h})$, $p$ cannot divide a non-zero integer smaller than itself. Therefore, no overlaps occur, verifying the collision axiom. □

By Theorem 3.3, these structural bounds ensure the convergence of the local deviations, so the distribution of twin prime tuples aligns with a Poisson point process.

## 5. Landau's First Problem (Fluctuating Dimension)

The transition from linear configurations to quadratic forms introduces an additional difficulty: dimensional fluctuation. To model the distribution of primes of the form $n^2 + 1$ (Landau's First Problem), the multi-residue sieve must exclude roots of $x^2 + 1 \equiv 0 \pmod{p}$. Unlike the linear case, the number of distinct roots is not constant but depends on the residue of $p$ modulo 4.

We resolve this by embedding the Legendre symbol directly into the density map, showing that fluctuating dimension is a special case of the same admissible geometry.

**Definition 5.1** (Quadratic Density Map [ক্ষ])**.** For the $n^2 + 1$ sieve, the density map $\kappa(p)$ counts the roots of $x^2 \equiv -1 \pmod{p}$. Utilizing the Legendre symbol for odd primes, we define:

$$\kappa(p) = \begin{cases} 1 & \text{if } p = 2 \\ 1 + \left(\frac{-1}{p}\right) & \text{if } p > 2 \end{cases}$$

This gives $\kappa(p) = 2$ for primes $p \equiv 1 \pmod{4}$ and $\kappa(p) = 0$ for primes $p \equiv 3 \pmod{4}$, deactivating the sieve dimension for half the prime population.

**Lemma 5.2** (Quadratic Admissibility)**.** *The fluctuating density map* $\kappa(p) = 1 + \left(\frac{-1}{p}\right)$ *satisfies the boundedness and non-degeneracy axioms of a Poisson-admissible sieve.*

*Proof.* For any odd prime, the Legendre symbol is restricted to $\{-1, 1\}$. Therefore, the density map is bounded by $1 + 1 = 2$, giving a uniform bound $B = 2$.

For non-degeneracy, $\kappa(2) = 1 < 2$. For $p \equiv 1 \pmod 4$, $p \geq 5$, yielding $\kappa(p) = 2 < 5 \leq p$. For $p \equiv 3 \pmod 4$, $p \geq 3$, yielding $\kappa(p) = 0 < 3 \leq p$. In all cases, the survival probability factors $1 - \kappa(p)/p$ are positive. □

To prove that this fluctuating sieve converges to a Poisson limit, we must constrain the variance of the active primes ($p \equiv 1 \pmod 4$). The intersections of quadratic roots must be bounded by finite Diophantine constants.

Because any prime of the form $x^2 + 1$ strictly greater than 2 requires an even base coordinate $x$, any valid shifted configuration $x + h_i$ must inherently preserve this parity. We therefore restrict our analysis to shifts composed of even integers, ensuring the $k$-tuple remains algebraically capable of surviving the $p = 2$ baseline density filter.

**Definition 5.3** (Quadratic Collision Threshold [ঋ])**.** For a fixed $k$-tuple of distinct even integers $\mathbf{h} = \{h_1, \ldots, h_k\}$, we define the prime threshold $T(\mathbf{h})$ bounding the quadratic shift collisions as:

$$T(\mathbf{h}) = \max_{i \neq j} \left((h_i - h_j)^2 + 4\right)$$

**Lemma 5.4** (Finiteness of Quadratic Overlaps)**.** *For all primes* $p > T(\mathbf{h})$ *satisfying* $p \equiv 1 \pmod 4$*, the excluded residue classes generated by the shifts are disjoint.*

*Proof.* Let $\nu$ be a local root of $x^2 \equiv -1 \pmod p$. The excluded classes modulo $p$ generated by a target shift $h_i$ are $\nu - h_i$ and $-\nu - h_i$. A collision between the excluded classes of two distinct shifts $h_i$ and $h_j$ falls into two cases. A same-sign collision ($\nu - h_i \equiv \nu - h_j$ or $-\nu - h_i \equiv -\nu - h_j$) requires $p \mid (h_i - h_j)$, which is impossible for $p > T(\mathbf{h}) \geq (h_i - h_j)^2 + 4 > |h_i - h_j|$. A cross-sign collision ($\nu - h_i \equiv -\nu - h_j$) requires $2\nu \equiv h_i - h_j \pmod p$.

Squaring both sides of this last equivalence yields $4\nu^2 \equiv (h_i - h_j)^2 \pmod p$. Because $\nu$ is a root of $x^2 \equiv -1 \pmod p$, we substitute $4\nu^2 \equiv -4 \pmod p$, producing the Diophantine condition:

$$-4 \equiv (h_i - h_j)^2 \pmod p \implies p \mid \left((h_i - h_j)^2 + 4\right)$$

By Definition 5.3, the value $(h_i - h_j)^2 + 4$ is a positive integer constant, and $T(\mathbf{h})$ is its maximum across all shift pairings. For any prime $p > T(\mathbf{h})$, this divisibility condition also fails, so no overlap of either type occurs. □

By embedding the Legendre symbol into the Poisson-admissible sieve framework, Lemma 5.4 shows that Landau's First Problem has the same collision structure as the twin prime sieve. Both are constrained by finite collision thresholds, ensuring they fall within the scope of the Grand Convergence Theorem.

## 6. Concluding Remarks

The unified framework presented here differs from traditional heuristic approaches. Models based on the Hardy–Littlewood conjecture [5] or the generalized Bateman–Horn heuristic [1] typically treat the occurrence of prime values in polynomials as

independent, pseudo-random events. By contrast, our multi-residue sieve derives its statistical behavior from the deterministic intersections of modular residue classes.

By abstracting the sieve into the Poisson-admissible axioms, we have separated the fluctuating dimension of quadratic forms from the underlying geometric correlation. The localized variance—whether tracking the constant roots of $x(x+2)$ or the Legendre-dependent roots of $x^2+1$—is finite, bounded by elementary Diophantine estimates.

Poisson-distributed spacings are therefore not an empirical coincidence but a consequence of bounded-variance multi-residue sieves. The local geometric errors vanish at rate $O(1/p^2)$, driving the $k$-level correlations to converge to the volume of the test box.

Framing prime-representing polynomials through the lens of bounded-variance geometric correlation provides a structured, verifiable foundation that replaces the assumption of prime randomness with the deterministic structure of modular arithmetic.

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